

While not all Series are Geometric Series, we can Pretend they are similar enough to a geometric Series to evaluate Convergence.

A geometric Series is of the form $\sum ar^n$ where r is the ratio of two consecutive terms. We use this idea of a ratio to get the following test:

Ratio Test

Let $\sum_{n=1}^{\infty} a_n$ be the series in question.

1) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (thus convergent)

2) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent

3) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$, you cannot conclude anything.

Pf/ we want prove part 1 because its annoying and makes me sad. Part 2 is nice though.

2) If $|a_{n+1}/a_n| \rightarrow L < 1$ as $n \rightarrow \infty$, then there is some N s.t.

$$\left| \frac{a_{n+1}}{a_n} \right| > 1 \quad \text{for } n \geq N$$

$$\text{So } |a_{n+1}| > |a_n| \quad \text{for } n \geq N$$

So $\lim_{n \rightarrow \infty} a_n \neq 0$ and $\sum a_n$ diverges by divergence test.

Ex: $\sum (-1)^n \frac{n^3}{3^n}$

$$a_{n+1} = (-1)^{n+1} \frac{(n+1)^3}{3^{n+1}}, \quad a_n = (-1)^n \frac{n^3}{3^n}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{\frac{(n+1)^3}{3^{n+1}}}{\frac{n^3}{3^n}} = \lim_{n \rightarrow \infty} \frac{3^n}{3^{n+1}} \cdot \frac{(n+1)^3}{n^3} = \frac{1}{3}$$

So $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{3} < 1$, thus $\sum (-1)^n \frac{n^3}{3^n}$ is absolutely Convergent ^{thus Convergent} via Ratio Test

Ex: $\sum \frac{n^n}{n!}$

$$a_n = \frac{n^n}{n!}, \quad a_{n+1} = \frac{(n+1)^{n+1}}{(n+1)!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{\frac{(n+1)^{n+1}}{(n+1)!}}{\frac{n^n}{n!}} = \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n}$$

$\hookrightarrow (n+1)! = (n+1)(n!)$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)^{n+1}}{n+1} \cdot \frac{1}{n^n} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

$S_c \sum_{n=1}^{\infty} \frac{n^n}{n!}$ diverges by ratio test.

To deal with a specific issue in Series, we have another test for any series with an n^{th} power (i.e. c^n). It is similar

Root Test:

Take the Series $\sum a_n$

1) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L < 1$, then the Series $\sum_{n=1}^{\infty} a_n$ is absolutely Convergent

2) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent.

3) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$, you cannot determine anything.

Ex:

$$\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$$

$$a_n = \left(\frac{2n+3}{3n+2} \right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{2n+3}{3n+2} \right)^n} = \lim_{n \rightarrow \infty} \frac{2n+3}{3n+2} = \frac{2}{3} < 1$$

So, $\sum \left(\frac{2n+3}{3n+2} \right)$ is absolutely Convergent.

Ex:

$$\sum_{n=1}^{\infty} \left(\frac{n+1}{n}\right)^n$$

$$a_n = \left(\frac{n+1}{n}\right)^n$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{n+1}{n}\right)^n} = \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$$

So root test doesn't work.

$$\text{Notice } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

So the series diverges by divergence test.

Strategy for testing Series

If you don't know what test to use, try this order

1. Divergence Test
2. p -Series
3. Geometric Series
4. Comparison Test / Limit Comparison Test
5. Alternating Series Test
6. Ratio Test
7. Root Test
8. Integral Test

Note: If the summand looks like a simple integral, you can just use Integral test first.